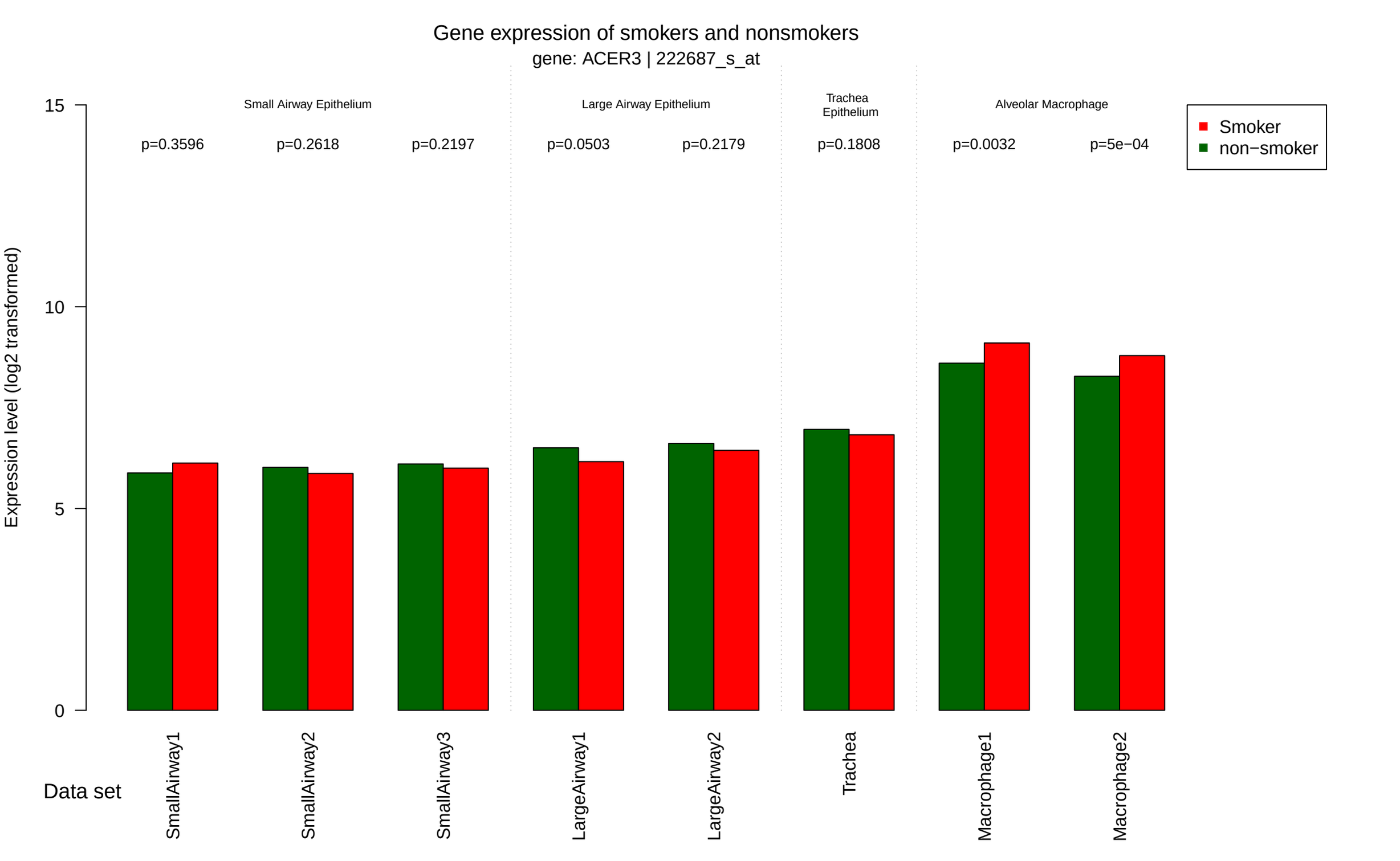
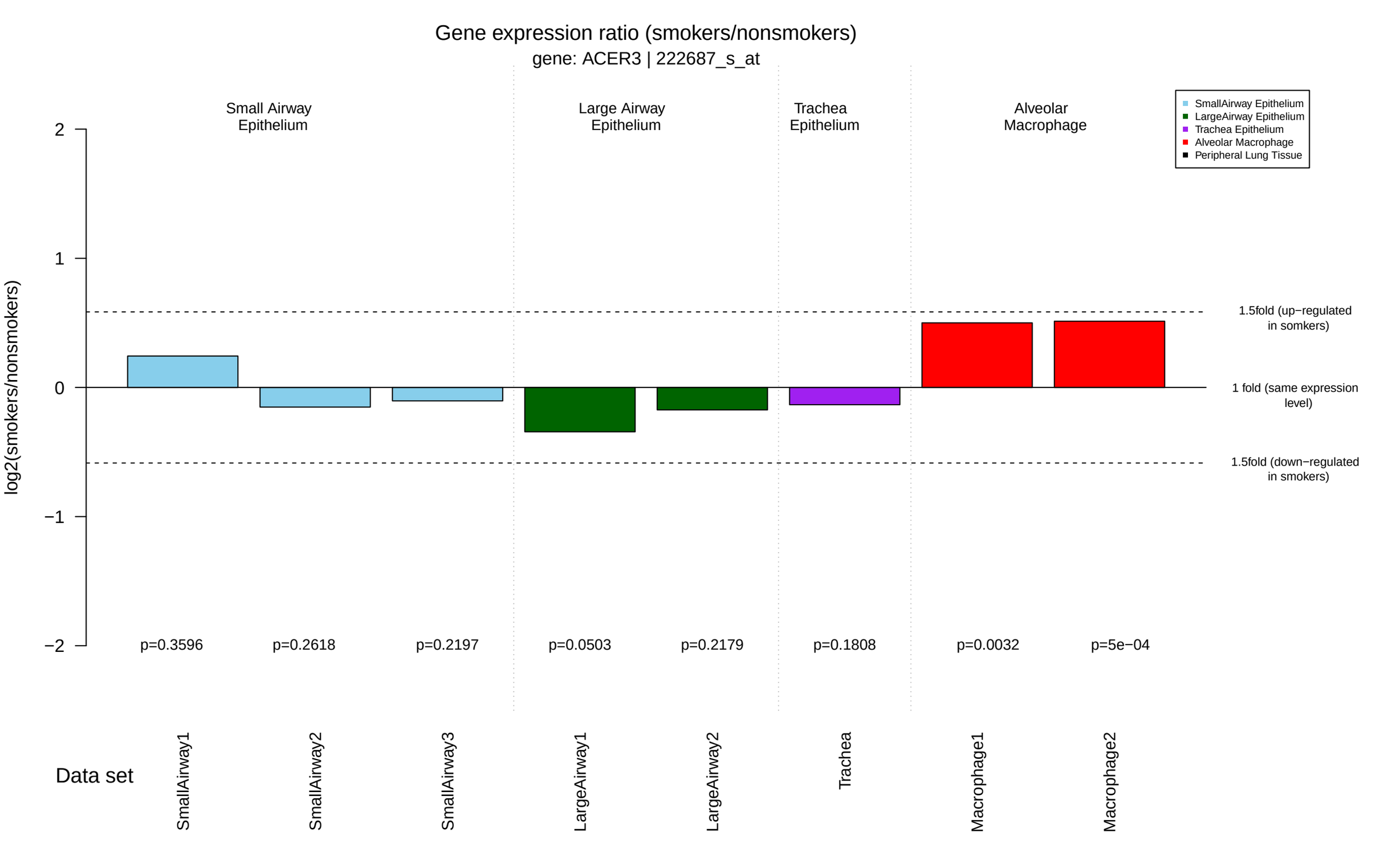
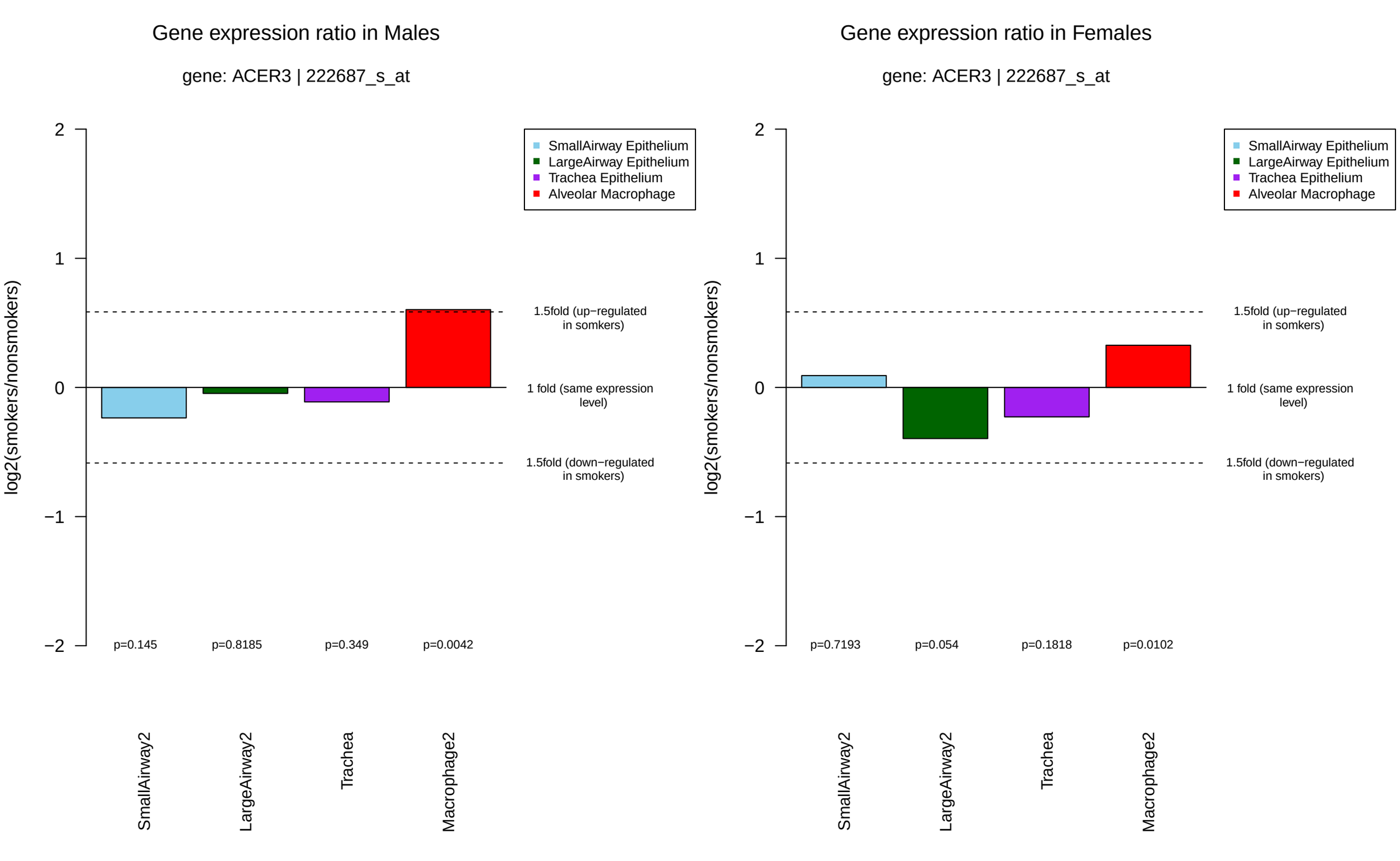
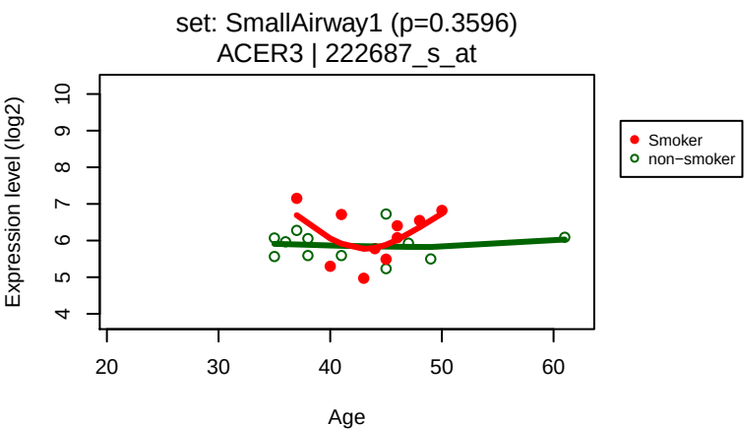
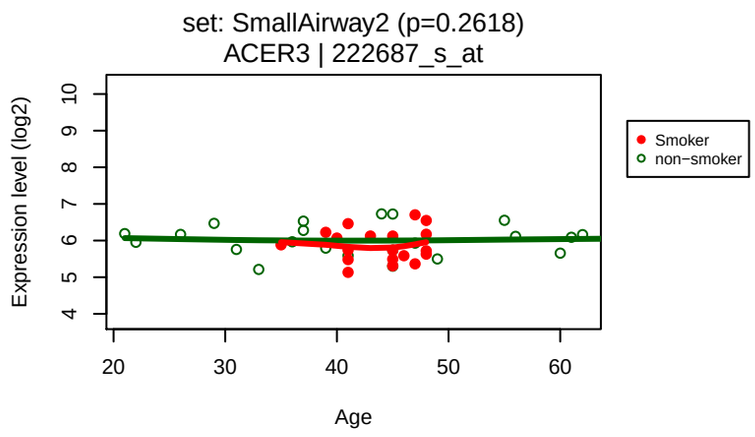
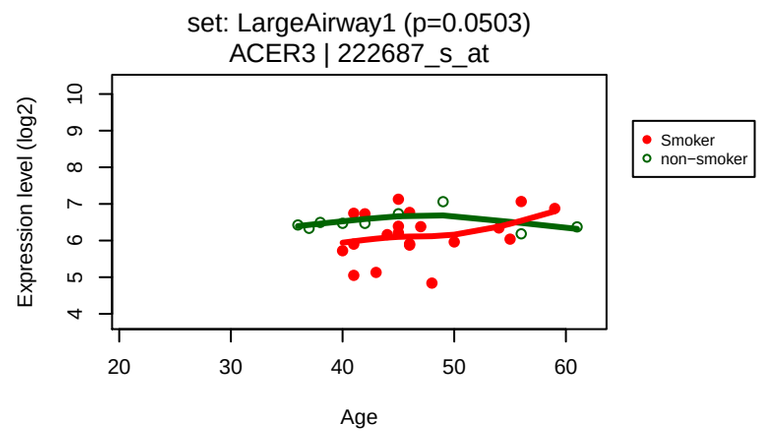

Gene expression of smokers and nonsmokers
gene: ACER3 | 222687_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3596
p=0.2618
p=0.2197
p=0.0503
p=0.2179
p=0.1808
p=0.0032
p=5e−04
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

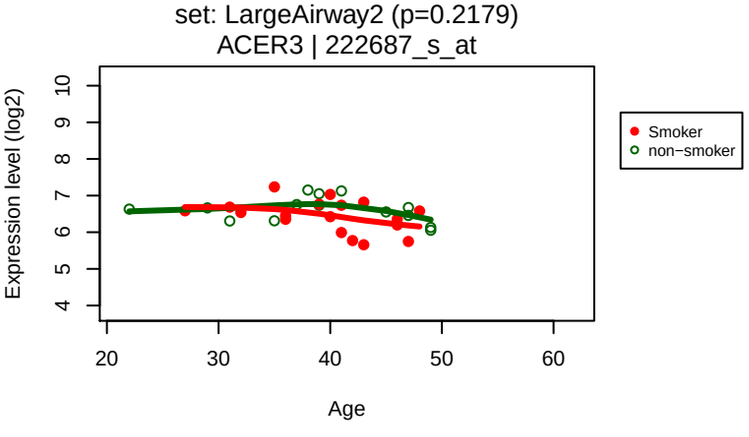

Gene expression ratio (smokers/nonsmokers)
gene: ACER3 | 222687_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3596
p=0.2618
p=0.2197
p=0.0503
p=0.2179
p=0.1808
p=0.0032
p=5e−04
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

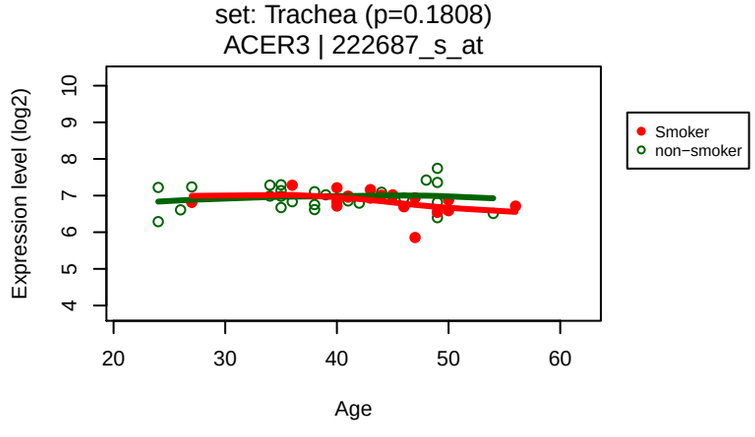

Gene expression ratio in Males
gene: ACER3 | 222687_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.145
p=0.8185
p=0.349
p=0.0042
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACER3 | 222687_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7193
p=0.054
p=0.1818
p=0.0102
SmallAirway2
LargeAirway2
Trachea
Macrophage2

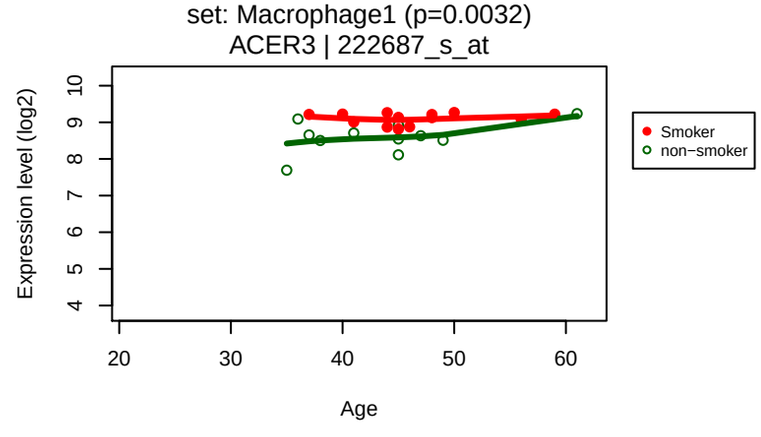

set: SmallAirway1 (p=0.3596)
ACER3 | 222687_s_at
Expression level (log2)
Age
Smoker
non−smoker

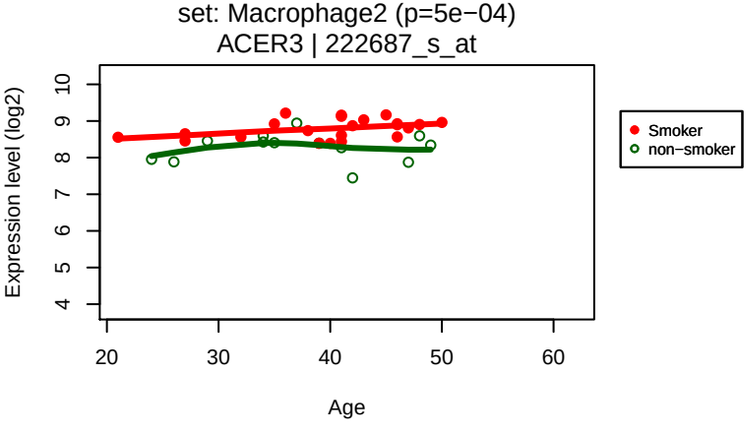

set: SmallAirway2 (p=0.2618)
ACER3 | 222687_s_at
Expression level (log2)
Age
Smoker
non−smoker

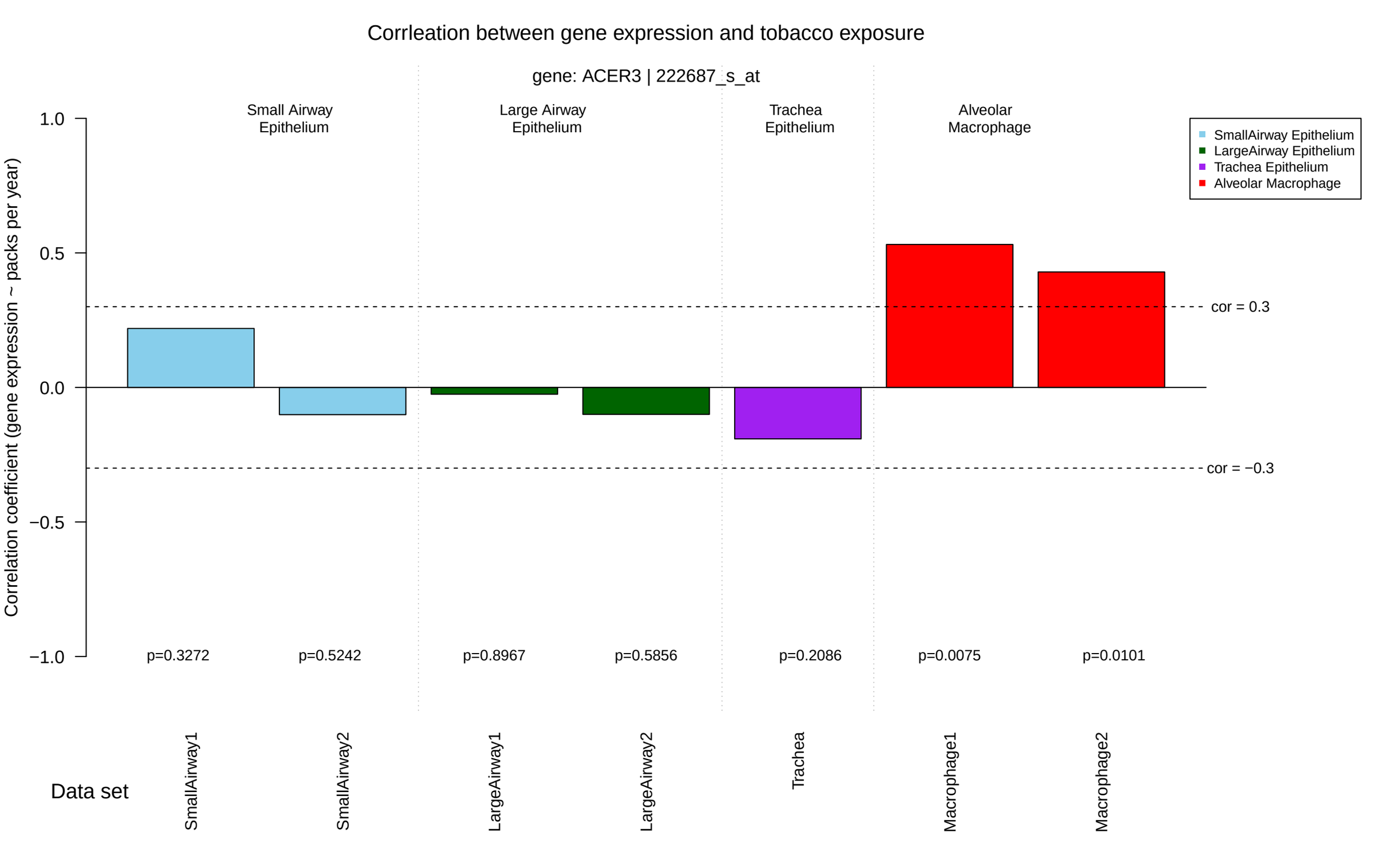

set: LargeAirway1 (p=0.0503)
ACER3 | 222687_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2179)
ACER3 | 222687_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1808)
ACER3 | 222687_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0032)
ACER3 | 222687_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=5e−04)
ACER3 | 222687_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACER3 | 222687_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3272
p=0.5242
p=0.8967
p=0.5856
p=0.2086
p=0.0075
p=0.0101
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2